Appendix A.

Analytical Constructs and Measurement Scales/ Online Supplementary Material

A.1 Power User Index (PUI)

To support comparative interpretation of AI adoption intensity among participants, the study employed an exploratory analytical construct referred to as the **Power User Index (PUI)**. The PUI was designed to synthesize qualitative interview data into a standardized indicator reflecting the degree to which AI tools are integrated into everyday work practices. The index serves an **interpretive and comparative purpose** within the studied organizational context and is not intended as a validated psychometric measurement instrument.

A.1.1 PUI Dimensions

The PUI integrates four dimensions capturing complementary aspects of workplace AI adoption:

1. **Proficiency (35%)**
   Proficiency reflects the participant's demonstrated AI-related capabilities and was assessed based on three qualitative indicators identified during interviews:
   - ability to create or customize AI-based solutions,
   - understanding of AI mechanisms and tool functionality,
   - evidence of an AI-first mindset in problem-solving. Each indicator was coded dichotomously (present/absent), resulting in a proficiency score ranging from 0 to 3.
2. **Intensity of Use (30%)**
   Intensity captures the extent to which AI tools are embedded in daily work routines. It was operationalized as self-reported average weekly time spent using AI tools for professional tasks.
3. **Breadth of Use (25%)**
   Breadth represents the diversity of AI tools utilized by a participant. Tool usage was categorized into five functional groups: conversational AI, multipurpose workplace tools, application creators, agentic systems, and coding assistants. The breadth score reflects the number of distinct categories used, regardless of specific vendors.
4. **Adoption Orientation (Sentiment) (10%)**
   This dimension captures participants' expressed orientation toward AI adop-

tion, based on qualitative indicators such as proactive learning behavior, experimentation, and advocacy for AI usage. It was coded as a binary variable (enthusiastic vs. non-enthusiastic orientation).

The weighting scheme reflects a deliberate hierarchy emphasizing **actualized capability and behavioral integration** over attitudinal factors. Proficiency and intensity jointly account for the majority of the index weight, ensuring that the PUI primarily reflects current operational AI use rather than future potential or interest.

A.1.2 Index Construction and Classification

Prior to aggregation, all component scores were normalized to a 0–100 scale to ensure comparability across dimensions. The weighted components were then aggregated into a composite PUI score.

To support analytical grouping, participants were classified into four adoption profiles (Non-Users, Casual Users, Active Users, Power Users) using the **Jenks Natural Breaks optimization method**, which maximizes between-group variance while minimizing within-group variance. The resulting thresholds are reported in Table 2 in the main text. Given the exploratory nature of the study and the limited sample size, the classification is intended solely for **pattern identification and interpretive comparison**.

A.2 AI Knowledge-Acquisition Channels and Usefulness Scale

To examine how employees acquire AI-related knowledge and skills, participants were asked to evaluate selected **AI knowledge-acquisition channels** based on their personal experience. The analyzed channels included, among others: formal AI training, advanced AI courses, live learning sessions, peer discussions, hackathons and challenges, internal communication platforms, and newsletters.

Each channel was evaluated using a **five-point ordinal scale**:

- 0 – Not used
- 1 – Used, not useful
- 2 – Used, moderately useful
- 3 – Used, useful
- 4 – Used, most useful

To distinguish perceived usefulness from mere popularity, a **Weighted Mean Utility Index** was calculated for each channel. The index represents the average usefulness score computed exclusively for participants who reported having used a given channel.

By excluding non-users from the calculation, the index enables a more accurate comparison of the **perceived educational value** of different learning mechanisms.

A.3 Interpretive Scope

Both the Power User Index and the learning-channel usefulness scale were developed specifically for this exploratory study and are **context-dependent analytical constructs**. They are intended to support interpretation of observed patterns in workplace AI learning and adoption rather than to provide universally generalizable measurement instruments. Their primary purpose is to facilitate the discussion of educational implications for BPM-oriented organizations and AI capability development initiatives.

Appendix B.

Design Patterns for AI-Enabled BPM Education

This supplementary material presents a set of **design patterns for AI-enabled BPM education**, derived from the empirical findings reported in the main paper. The patterns translate observed learning behaviors, learner typologies, and institutional barriers into **actionable educational solutions** for BPM educators and program designers. The patterns are intended as **practice-oriented guidance**, not as prescriptive instructional models.

Pattern 1: Staged AI Capability Development

Problem
Learners exhibit heterogeneous levels of AI proficiency and adoption intensity. Uniform training programs fail to address the needs of both novice and advanced users, leading to disengagement or cognitive overload.

Context
Observed AI adoption follows a staged trajectory, where early learning experiences strongly influence long-term adoption ("adoption armor").

Solution
Design AI-enabled BPM education as a **staged learning pathway**, progressing from basic AI-supported tasks (e.g., information retrieval, documentation support) toward advanced applications (e.g., process analysis, automation, agentic AI).

Educational Implications for BPM

- Align AI learning stages with BPM competency development (e.g., modeling → analysis → improvement).
- Emphasize early success experiences to reduce long-term adoption barriers.

Pattern 2: Hybrid Learning Ecosystems

Problem
Purely formal training or purely exploratory learning leads to suboptimal outcomes: either tool overload or fragmented knowledge.

Context
Empirical results show that the highest AI proficiency emerges when **structured learning**, **peer collaboration**, and **guided experimentation** are combined.

Solution
Design BPM education as a **hybrid learning ecosystem** combining:

- structured instruction (formalists),
- collaborative learning spaces (collaborators),
- controlled experimentation environments (experimenters).

Educational Implications for BPM

- Integrate workshops, peer sessions, and sandbox exercises into BPM courses.
- Treat experimentation as a supported activity, not an informal add-on.

Pattern 3: Learning-Channel Efficiency over Popularity

Problem
Highly popular learning channels (e.g., newsletters) show low educational effectiveness, while effective formats (e.g., hackathons) suffer from low participation.

Context
The study reveals a systematic gap between **adoption** and **educational utility** of learning mechanisms.

Solution
Prioritize learning channels based on **educational efficiency**, not ease of deployment. Actively promote high-impact formats and reduce overreliance on passive information dissemination.

Educational Implications for BPM

- Design curricula around interactive formats even if participation must be actively facilitated.
- Use passive channels as complementary, not primary, learning mechanisms.

Pattern 4: Socially Embedded BPM–AI Learning

Problem
Individual learning approaches fail to leverage organizational knowledge and often result in isolated AI usage practices.

Context
Learners relying on peer collaboration and informal knowledge exchange achieve stable and sustainable AI adoption.

Solution
Embed AI learning within **communities of practice**, peer mentoring, and shared BPM problem-solving activities.

Educational Implications for BPM

- Encourage collaborative BPM assignments supported by AI tools.
- Design learning activities that require knowledge sharing and joint reflection.

Pattern 5: Integrating AI Skills with Process Knowledge

Problem
Technologically proficient learners may lack BPM domain understanding, while experienced BPM professionals may resist AI adoption.

Context
The observed tenure gap creates a risk of technology-driven rather than process-driven BPM evolution.

Solution
Design learning activities that explicitly **connect AI usage with BPM concepts**, process models, and organizational context.

Educational Implications for BPM

- Frame AI as a process-supporting capability, not a standalone technology.
- Use BPM cases and real organizational processes as learning anchors.

Pattern 6: Time-Aware Learning Design

Problem
Learners perceive AI learning as valuable but report insufficient time due to operational pressure (“time–expectation paradox”).

Context
Learning initiatives competing with operational tasks are systematically deprioritized.

Solution
Integrate AI learning directly into BPM activities and workflows rather than treating it as an extra task.

Educational Implications for BPM

- Design short, task-embedded learning units.
- Align learning objectives with real BPM deliverables.

Summary for BPM Educators

The presented design patterns emphasize that effective AI-enabled BPM education requires:

- differentiation by learner profile,
- integration of AI with BPM tasks,
- prioritization of active and social learning,
- and organizationally realistic learning designs.

These patterns complement the empirical findings of the main paper and provide actionable guidance for BPM educators seeking to design sustainable AI capability development initiatives.